

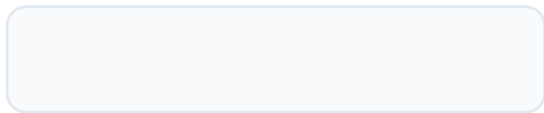
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Differential Cryptanalysis and Its Enhancements

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Differential Cryptanalysis (History)

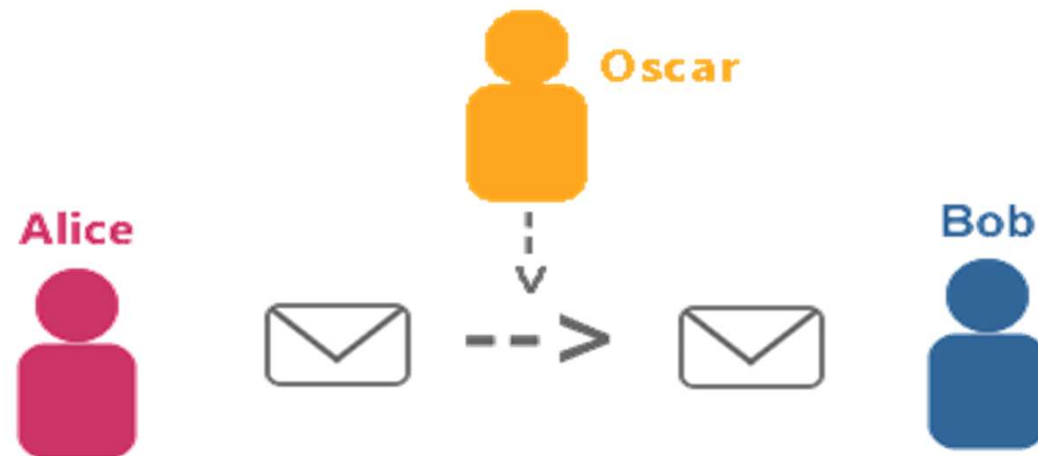
1.1990 — Eli Biham and Adi Shamir publish differential cryptanalysis as a general method for attacking block ciphers.

2.The attack was applied to DES.

3.1994 —IBM disclosed that they had discovered the technique in 1974 during DES design. The S-boxes were modified to resist it.

1.Post-1990 — Resistance to differential cryptanalysis becomes a standard design requirement. The wide trail strategy is developed as a formal defense.

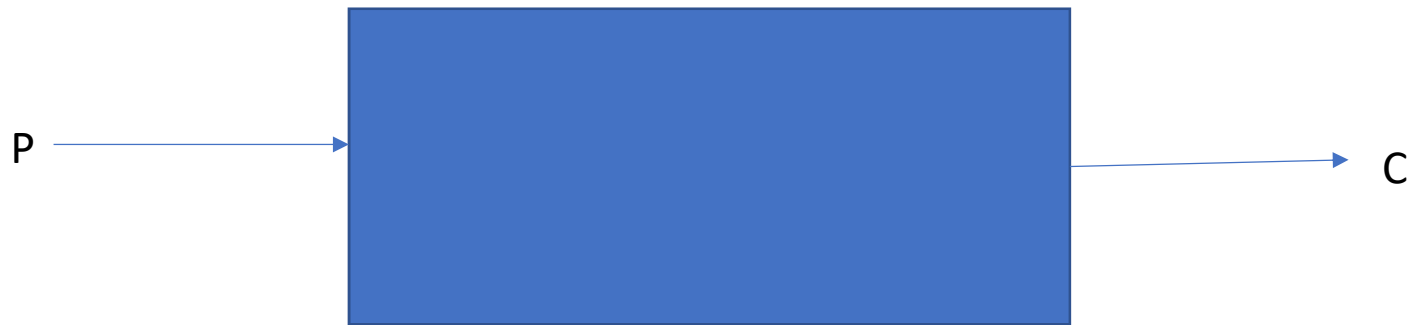
Background



Attack Type: Chosen Plaintext Attack

Goal of attacker: To find non-randomness in the ciphertext

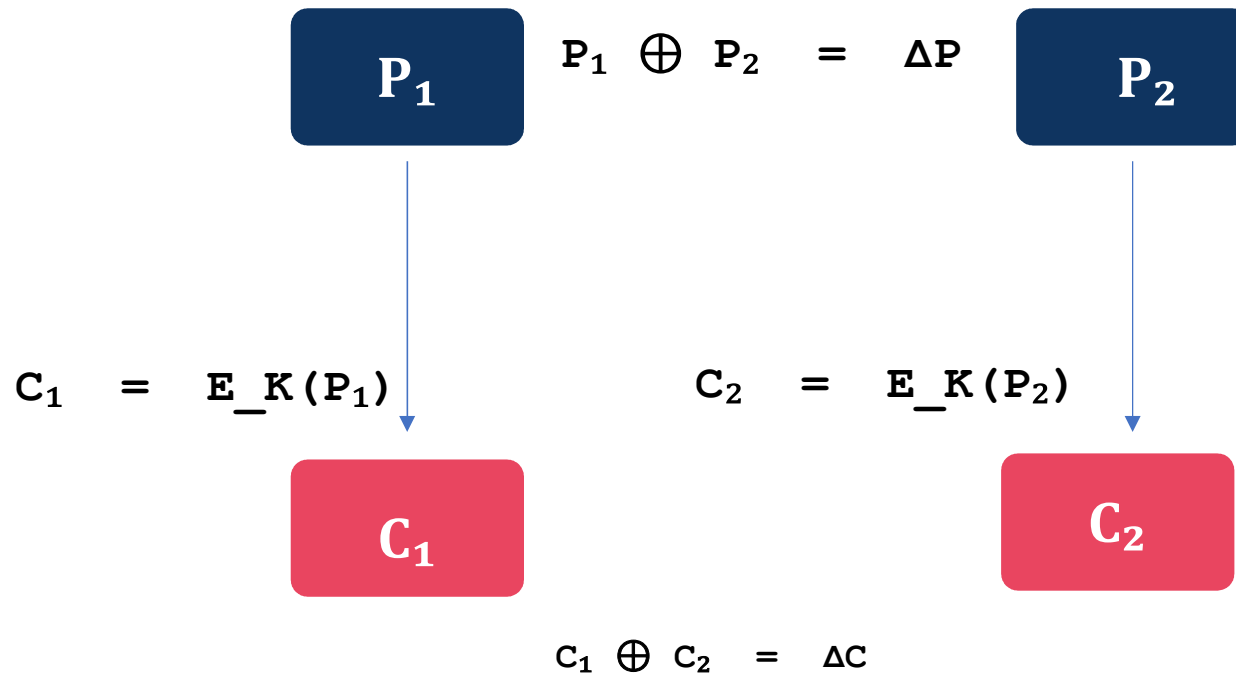
Motivation: (Identifying relation between input and output)



Motivation: (DES structure)

1. Round Key Addition in the beginning. (Secret)
2. Therefore, to identify something that is not dependent on key.

The Formal Structure:



Why XOR? (Significance for DES)

$$\begin{aligned} C_1 &= E_K(P_1) = S(P_1 \oplus K) \\ C_2 &= E_K(P_2) = S(P_2 \oplus K) \end{aligned}$$

Input difference to S-box:

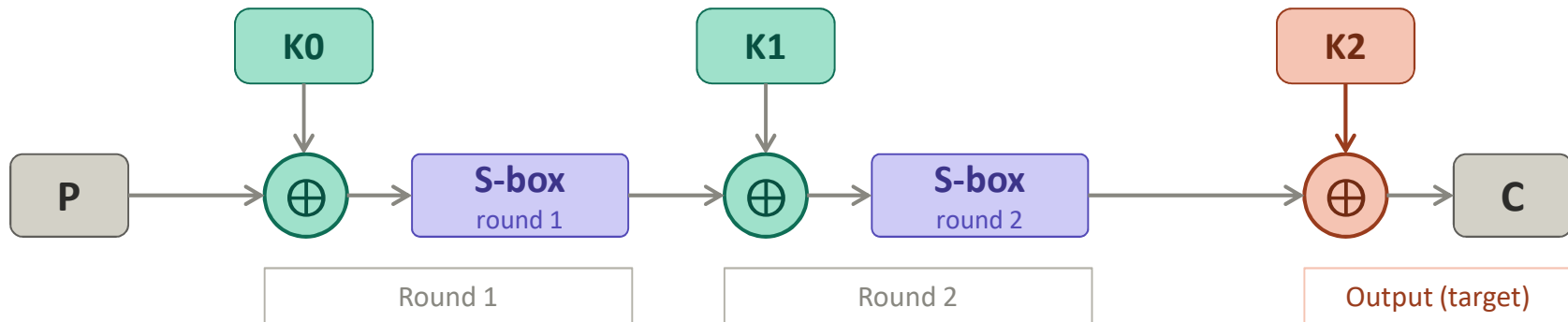
$$(P_1 \oplus K) \oplus (P_2 \oplus K) = P_1 \oplus P_2 = \Delta P$$

Key insight

The same key K is XORed into both. When you XOR the two inputs, K disappears. The input difference to the S-box depends only on ΔP — not on K .

Toy S-box Cipher — 4-bit Block, 2 Rounds

Keys K0, K1 drop in from above. K2 is the attack target (coral).



x	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
S(x)	E	4	D	1	2	F	B	8	3	A	6	C	5	9	0	7

Difference Distribution Table (DDT)

For every possible input difference Δ_{in} — how many input pairs produce each output difference Δ_{out} ?

1

Fix Δ_{in}

Choose an input difference,
e.g. $\Delta_{in} = 0011$

2

Enumerate pairs

For all $x \in \{0, \dots, 15\}$, compute
 $x' = x \oplus \Delta_{in}$

3

Compute Δ_{out}

Look up $S(x)$ and $S(x')$,
compute $S(x) \oplus S(x')$

4

Count

Tally how often each Δ_{out}
appears

5

Repeat

Do this for all 15 nonzero Δ_{in}
values

DDT row sum = 16
(for a 4-bit S-box)
Each row must account for all 16
input pairs.

DDT — Heys S-box (4-bit), values out of 16

Star differential: $\Delta B \rightarrow \Delta 2$, count = $8/16 = 1/2$ (coral). Purple = 6, teal = 4, white = 2, gray = 0.

$\Delta in \backslash \Delta out$	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	2	0	0	0	2	0	2	4	0	4	2	0	0
2	0	0	0	2	0	6	2	2	0	2	0	0	0	0	2	0
3	0	0	2	0	2	0	0	0	0	4	2	0	2	0	0	4
4	0	0	0	2	0	0	6	0	0	2	0	4	2	0	0	0
5	0	4	0	0	0	2	2	0	0	0	4	0	2	0	0	2
6	0	0	0	4	0	4	0	0	0	0	0	0	2	2	2	2
7	0	0	2	2	2	0	2	0	0	2	2	0	0	0	0	4
8	0	0	0	0	0	0	2	2	0	0	0	4	0	4	2	2
9	0	2	0	0	2	0	0	4	2	0	2	2	2	0	0	0
A	0	2	2	0	0	0	0	0	6	0	0	2	0	0	4	0
B	0	0	8	0	0	2	0	2	0	0	0	0	0	2	0	2
C	0	2	0	0	2	2	2	0	0	0	0	2	0	6	0	0
D	0	4	0	0	0	0	0	4	2	0	2	0	2	0	2	0
E	0	0	2	4	2	0	0	0	6	0	0	0	0	0	2	0
F	0	2	0	0	6	0	0	0	0	4	0	2	0	0	2	0

Key Recovery

1

Build a DDT for 1 rounds —

Leave the last round out intentionally.

2

Collect ciphertext pairs —

Encrypt many pairs with chosen ΔP . Collect (C_1, C_2) pairs.

3

Guess last-round subkey —

For each candidate K^* , partially decrypt C_1 and C_2 through the last round.

4

Check the characteristic —

Does the decrypted difference match expected Δ_{out} ?

5

Count votes —

Right key $\rightarrow$ signal. Wrong key $\rightarrow$ noise. Right key gets the most votes.

Wrong key guess $\rightarrow$ noise | Right key guess $\rightarrow$ signal | Same game as linear cryptanalysis

Drawback of DDT?

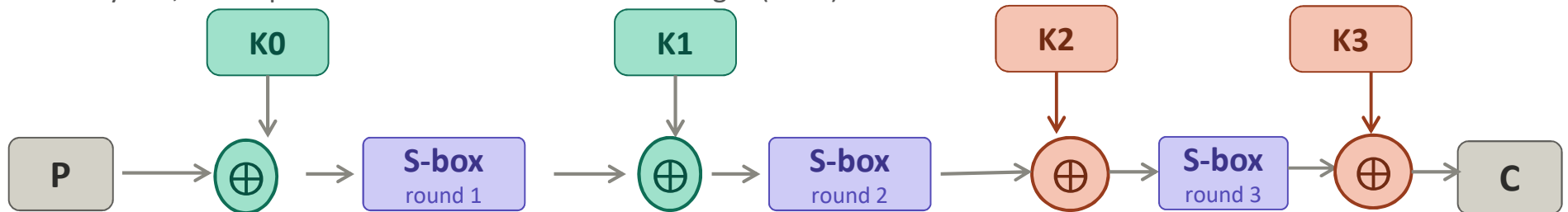
1. Size
2. Handling Multiple S boxes

Question:

1. How to handle multiple parallel S boxes
2. How to handle multiple serial S boxes
3. Sbox followed by permutation

Toy SPN Cipher — 4-bit Block, 2 Rounds

Keys K0, K1 drop in from above. K2 is the attack target (coral).



Teal = Round keys (XOR cancels in differences) Purple = S-box (non-linear) Coral = K2 (attack target)

x	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
S(x)	E	4	D	1	2	F	B	8	3	A	6	C	5	9	0	7

Differential Characteristic

A differential characteristic traces a specific input difference through every round of the cipher.



$$\text{Total probability} = p_1 \times p_2 \times p_3 = 1/8 \times 1/4 \times 1/8 = 1/256 = 2^{-8}$$

Characteristic

One specific sequence of differences round by round.
Fixed Δ_{in} and fixed Δ_{out} .

Differential

All pairs $(\Delta_{in}, \Delta_{out})$ with nonzero probability,
regardless of the path taken.

Truncated Differential

Definition

A differential where only some components of the states are specified.

The rest are left as ignored — they can take any value.

Example — 8-bit difference vector, split into two words:

Classical:

0110

|

0000

← exact value of both words fixed

Truncated:

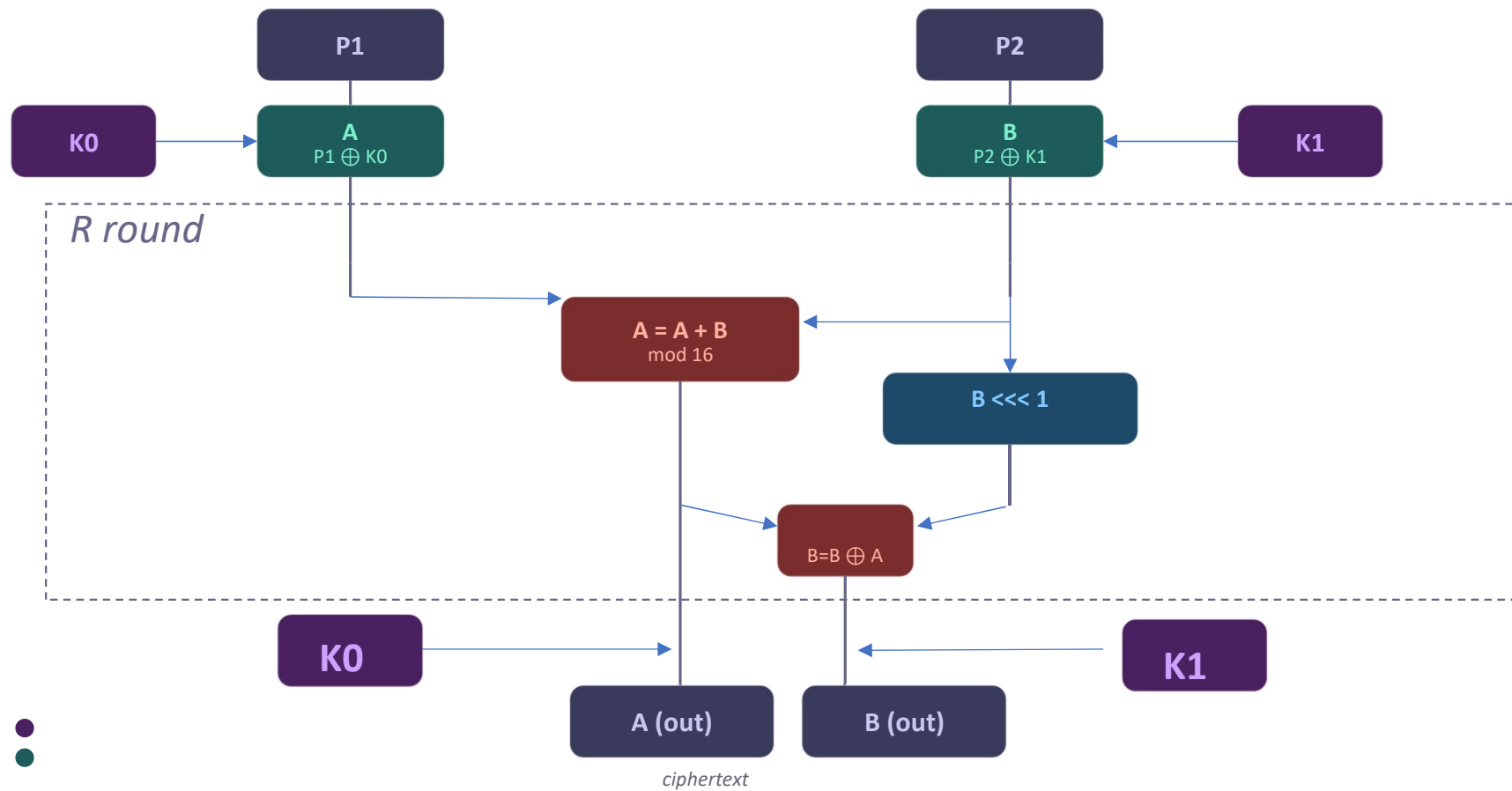
? ? ? ?

|

0000

← left word = don't care. Right = zero.

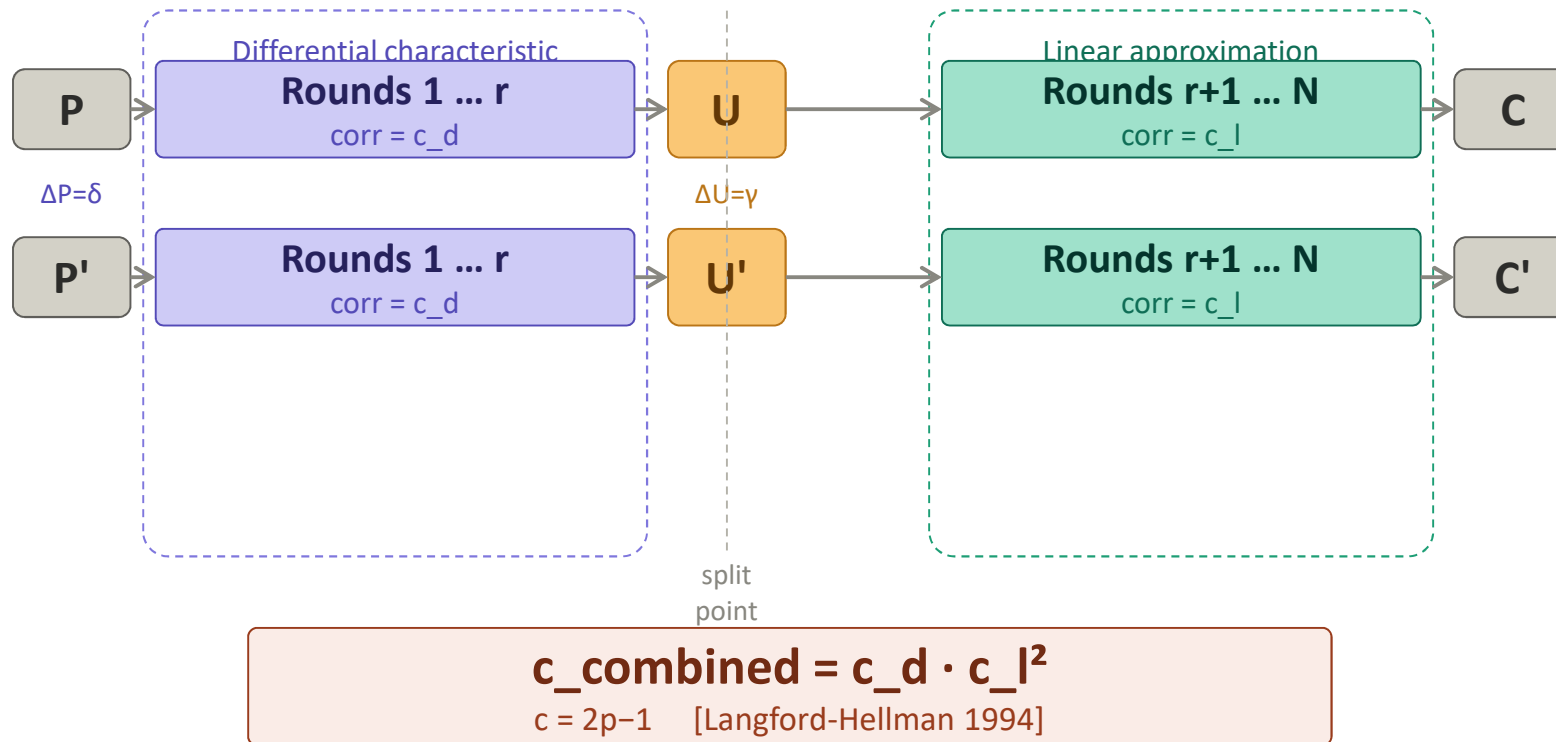
Toy ARX Cipher



Truncated Differential on Toy

1. Complement single bit
2. Check $P(C[i]=C'[i])$, (i-th bit)
3. Any probability other than 0.5 can be considered as a distinguisher.
(program)

Differential-Linear Attack



**THANK
YOU**